

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

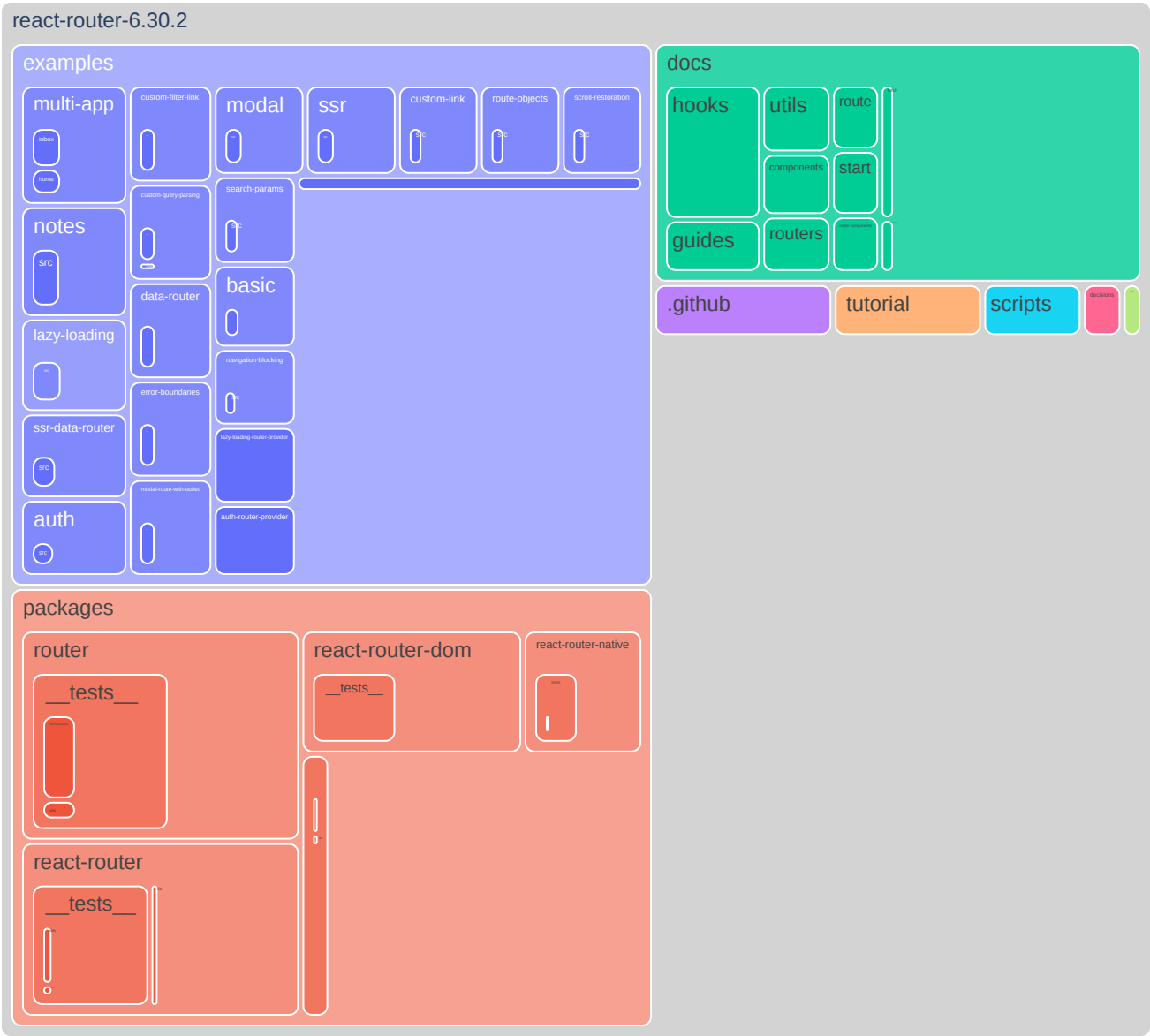
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	83.000000	83.000000	83.000000	83.000000	83.000000	83.000000
mean	25.253012	19.662651	130.060241	692.975904	1124.457831	694.783133
std	77.782662	46.320010	293.488245	333.698892	401.791667	334.133829
min	1.000000	2.000000	3.000000	49.000000	49.000000	49.000000
25%	4.000000	4.000000	11.000000	456.000000	854.500000	455.000000
50%	10.000000	6.000000	26.000000	841.000000	1210.000000	841.000000
75%	13.000000	13.000000	75.000000	940.000000	1533.000000	940.000000
max	633.000000	344.000000	1996.000000	1574.000000	1631.000000	1573.000000

	directoryPath	directoryParentPath	directoryName	fileCount	firstFile	mostF
0	react-router-6.30.2/examples/custom-query-parsing/types	react-router-6.30.2/examples/custom-query-parsing	types	1	react-router-6.30.2/examples/custom-query-parsing/types/jsurl.d.ts	
1	react-router-6.30.2/packages/react-router-dom-v5-compat/lib	react-router-6.30.2/packages/react-router-dom-v5-compat	lib	1	react-router-6.30.2/packages/react-router-dom-v5-compat/lib/components.tsx	
2	react-router-6.30.2/packages/react-router-dom/__tests__/polyfills	react-router-6.30.2/packages/react-router-dom/__tests__	polyfills	1	react-router-6.30.2/packages/react-router-dom/__tests__/polyfills/drop-FormData-submitter.ts	
3	react-router-6.30.2/packages/react-router/__tests___/_snapshots__	react-router-6.30.2/packages/react-router/__tests__	__snapshots__	1	react-router-6.30.2/packages/react-router/__tests___/_snapshots__/route-matching-test.tsx.snap	
4	react-router-6.30.2/patches	react-router-6.30.2	patches	1	react-router-6.30.2/patches/@changesets_get-dependents-graph@1.3.6.patch	
5	react-router-6.30.2/.changeset	react-router-6.30.2	.changeset	2	react-router-6.30.2/.changeset/README.md	
6	react-router-6.30.2/docs/upgrading	react-router-6.30.2/docs	upgrading	2	react-router-6.30.2/docs/upgrading/future.md	
7	react-router-6.30.2/examples/lazy-loading/src/pages	react-router-6.30.2/examples/lazy-loading/src	pages	2	react-router-6.30.2/examples/lazy-loading/src/pages/About.tsx	
8	react-router-6.30.2/.github/ISSUE_TEMPLATE	react-router-6.30.2/.github	ISSUE_TEMPLATE	3	react-router-6.30.2/.github/ISSUE_TEMPLATE/bug_report.yml	
9	react-router-6.30.2/examples/navigation-blocking/src	react-router-6.30.2/examples/navigation-blocking	src	3	react-router-6.30.2/examples/navigation-blocking/src/app.tsx	
10	react-router-6.30.2/packages/react-router-dom-v5-compat/__tests__	react-router-6.30.2/packages/react-router-dom-v5-compat	__tests__	3	react-router-6.30.2/packages/react-router-dom-v5-compat/__tests__/compat-router-test.tsx	
11	react-router-6.30.2/decisions	react-router-6.30.2	decisions	4	react-router-6.30.2/decisions/0001-use-blocker.md	
12	react-router-6.30.2/examples/basic/src	react-router-6.30.2/examples/basic	src	4	react-router-6.30.2/examples/basic/src/App.tsx	
13	react-router-6.30.2/examples/custom-link/src	react-router-6.30.2/examples/custom-link	src	4	react-router-6.30.2/examples/custom-link/src/App.tsx	
14	react-router-6.30.2/examples/custom-query-parsing/src	react-router-6.30.2/examples/custom-query-parsing	src	4	react-router-6.30.2/examples/custom-query-parsing/src/App.tsx	
15	react-router-6.30.2/examples/multi-app/home	react-router-6.30.2/examples/multi-app	home	4	react-router-6.30.2/examples/multi-app/home/App.jsx	
16	react-router-6.30.2/examples/route-objects/src	react-router-6.30.2/examples/route-objects	src	4	react-router-6.30.2/examples/route-objects/src/App.tsx	
17	react-router-6.30.2/examples/scroll-restoration/src	react-router-6.30.2/examples/scroll-restoration	src	4	react-router-6.30.2/examples/scroll-restoration/src/app.tsx	
18	react-router-6.30.2/examples/search-params/src	react-router-6.30.2/examples/search-params	src	4	react-router-6.30.2/examples/search-params/src/App.tsx	
19	react-router-6.30.2/packages/react-router-native/__tests___/_snapshots__	react-router-6.30.2/packages/react-router-native/__tests__	__snapshots__	4	react-router-6.30.2/packages/react-router-native/__tests___/_snapshots__/deep-linking-test.tsx.snap	
20	react-router-6.30.2/packages/react-router/lib	react-router-6.30.2/packages/react-router	lib	4	react-router-6.30.2/packages/react-router/lib/components.tsx	
21	react-router-6.30.2/packages/router/__tests___/utils	react-router-6.30.2/packages/router/__tests__	utils	4	react-router-6.30.2/packages/router/__tests___/utils/custom-matchers.ts	
22	react-router-6.30.2/tutorial/src	react-router-6.30.2/tutorial	src	4	react-router-6.30.2/tutorial/src/App.jsx	
23	react-router-6.30.2/.github/workflows	react-router-6.30.2/.github	workflows	5	react-router-6.30.2/.github/workflows/deduplicate-lock-file.yml	
24	react-router-6.30.2/docs/fetch	react-router-6.30.2/docs	fetch	5	react-router-6.30.2/docs/fetch/index.md	
25	react-router-6.30.2/examples/auth/src	react-router-6.30.2/examples/auth	src	5	react-router-6.30.2/examples/auth/src/App.tsx	
26	react-router-6.30.2/examples/custom-filter-link/src	react-router-6.30.2/examples/custom-filter-link	src	5	react-router-6.30.2/examples/custom-filter-link/src/App.tsx	
27	react-router-6.30.2/examples/data-router/src	react-router-6.30.2/examples/data-router	src	5	react-router-6.30.2/examples/data-router/src/app.tsx	
28	react-router-6.30.2/examples/error-boundaries/src	react-router-6.30.2/examples/error-boundaries	src	5	react-router-6.30.2/examples/error-boundaries/src/app.tsx	
29	react-router-6.30.2/examples/modal-route-with-outlet/src	react-router-6.30.2/examples/modal-route-with-outlet	src	5	react-router-6.30.2/examples/modal-route-with-outlet/src/App.tsx	

Number of files per directory

Directories and their file count



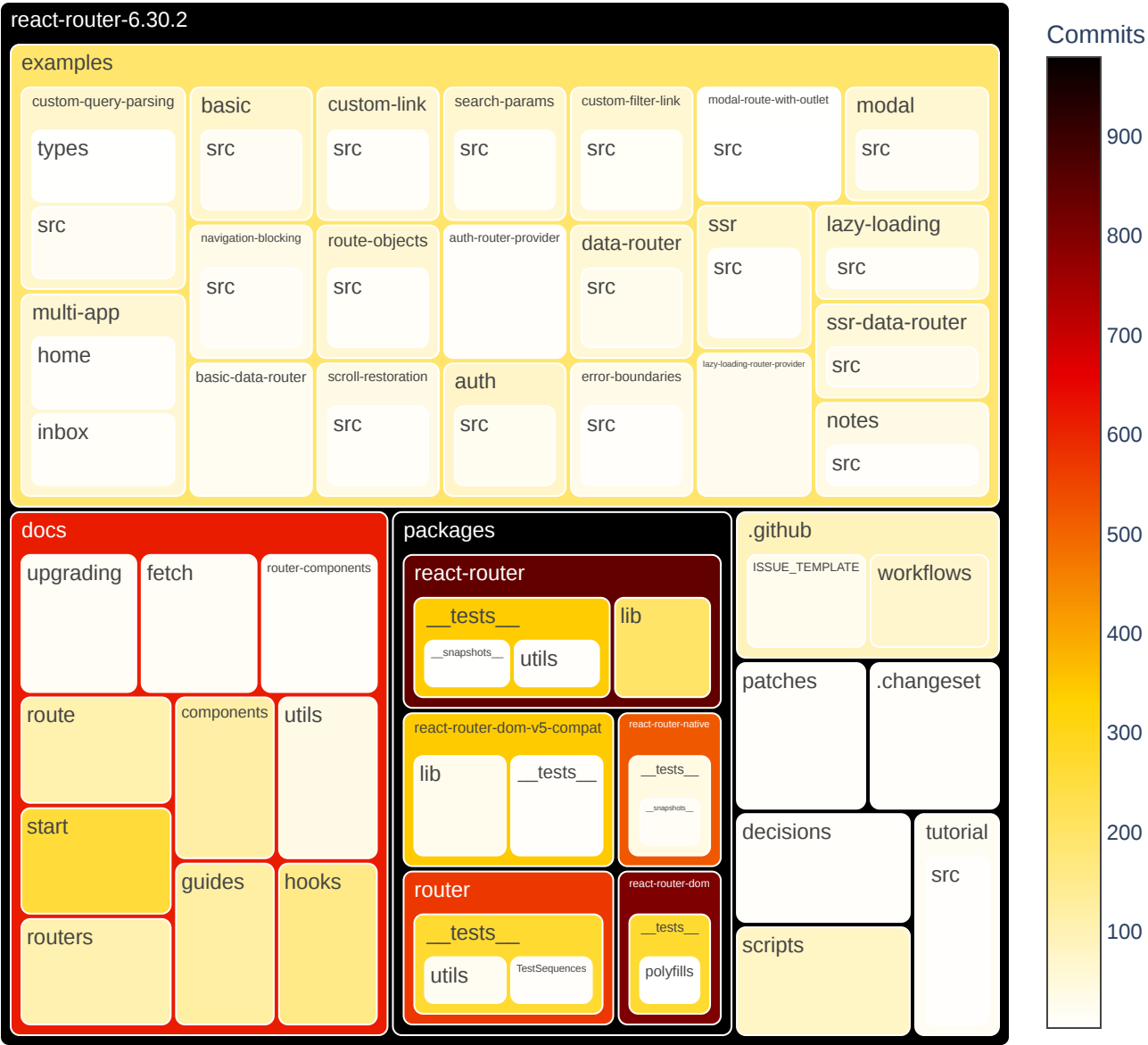
Most frequent file extension per directory

Most frequent file extension per directory



Number of commits per directory

Number of git commits



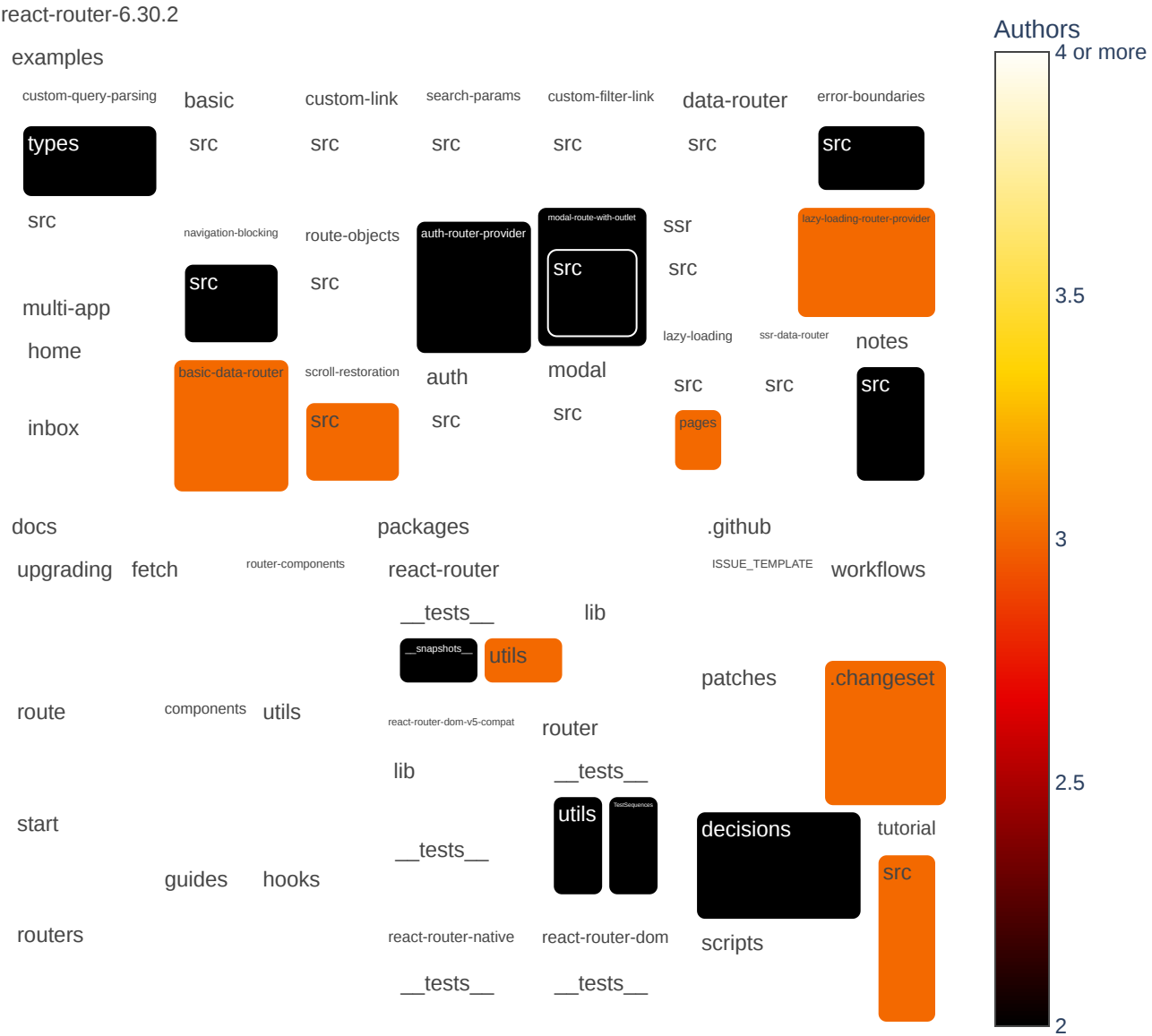
Number of distinct authors per directory

Number of distinct commit authors



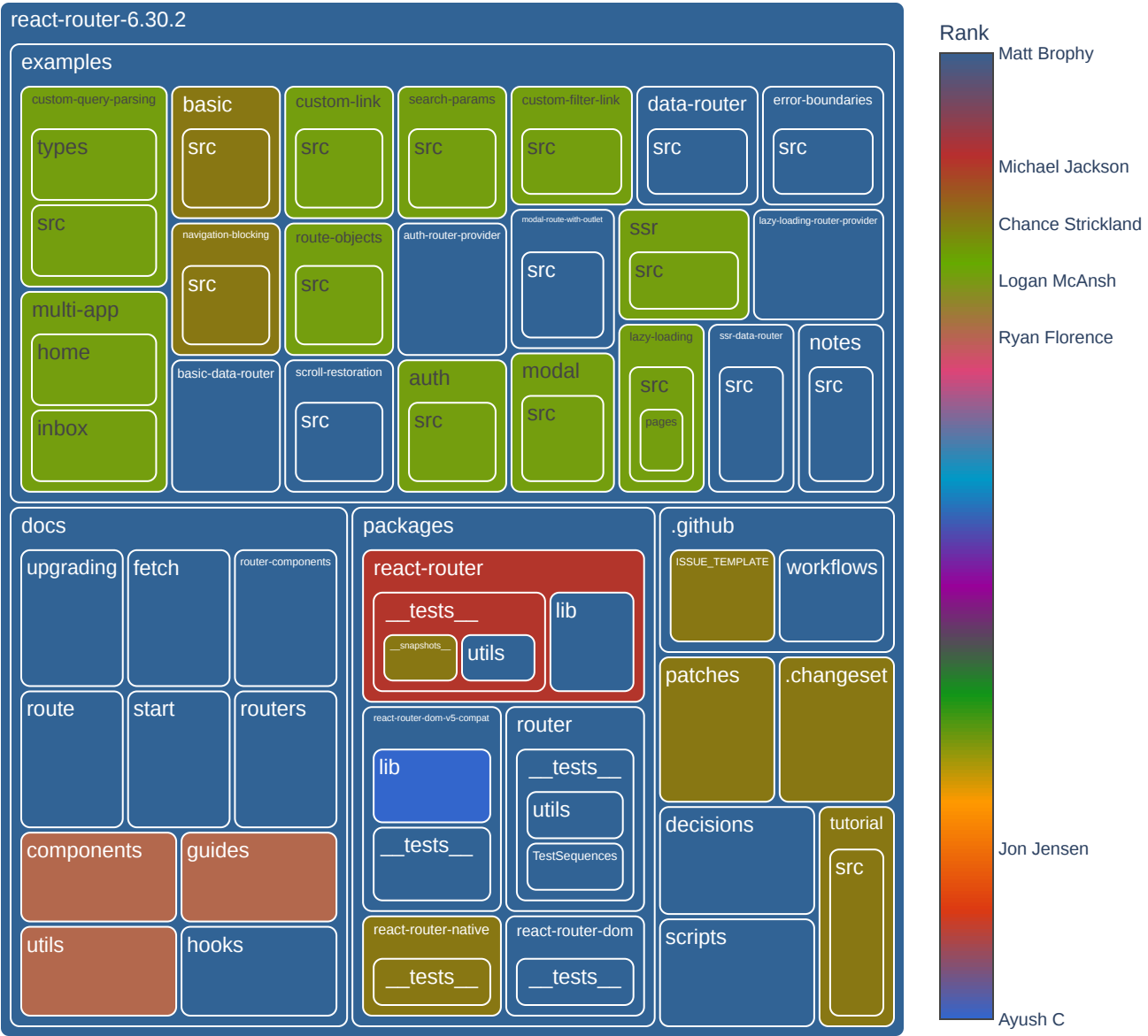
Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)



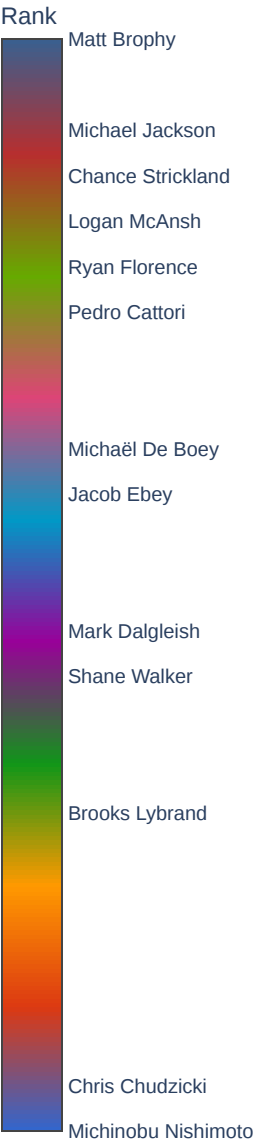
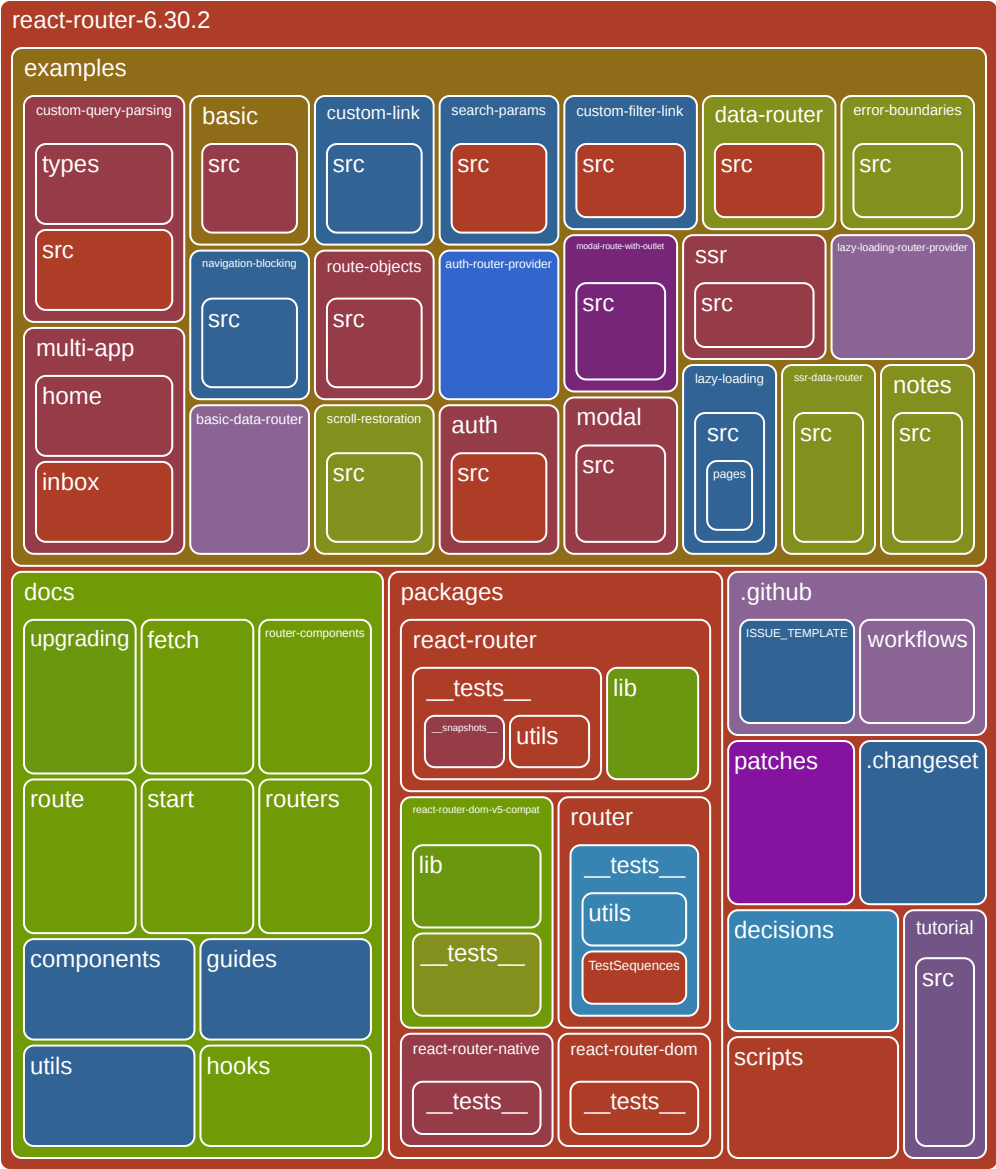
Main author per directory

Main authors with highest number of commits



Second author per directory

Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

react-router-6.30.2



Days since last commit per directory (ranked)

Rank of days since last commit

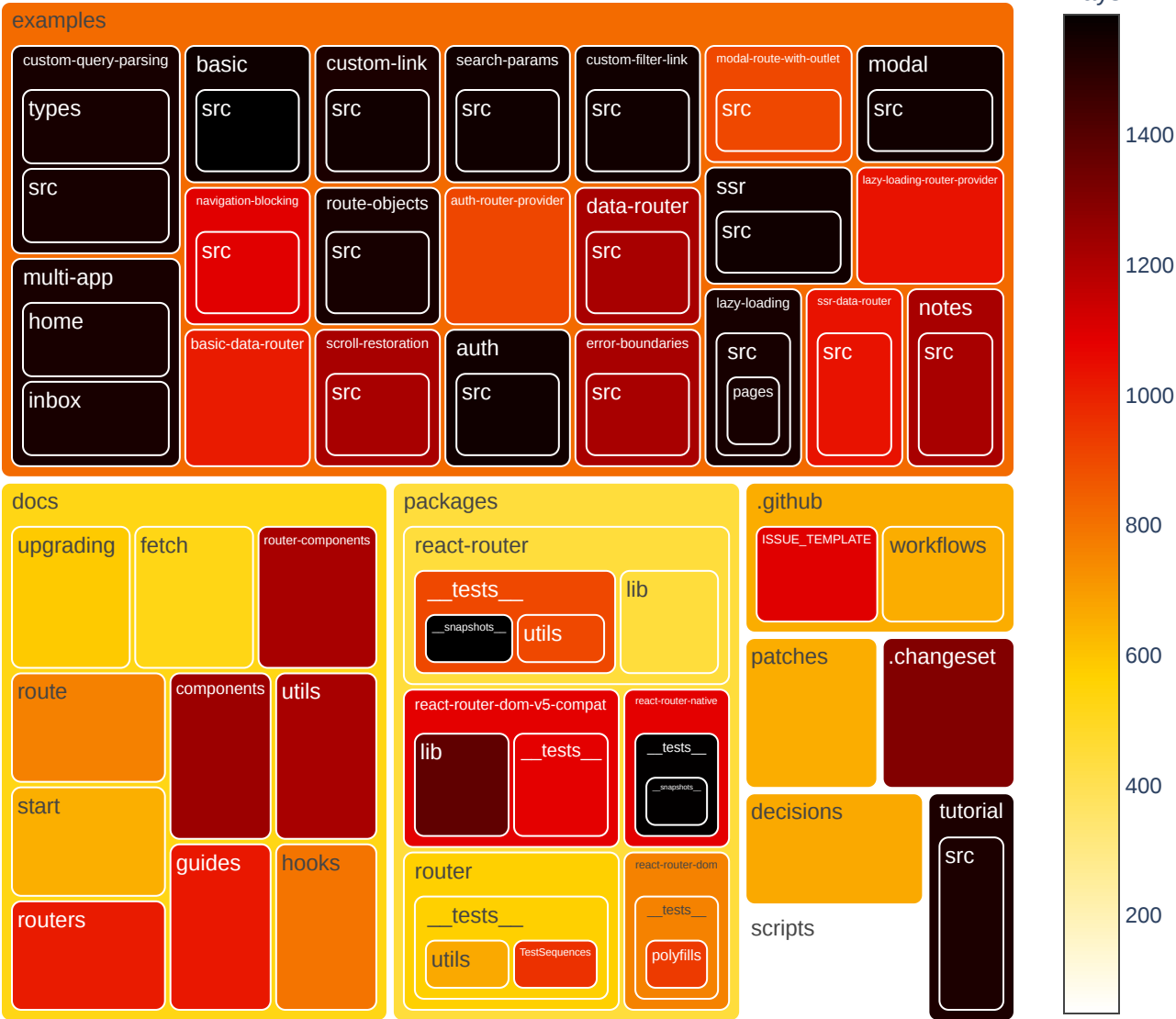
react-router-6.30.2



Days since last file creation per directory

Days since last file creation

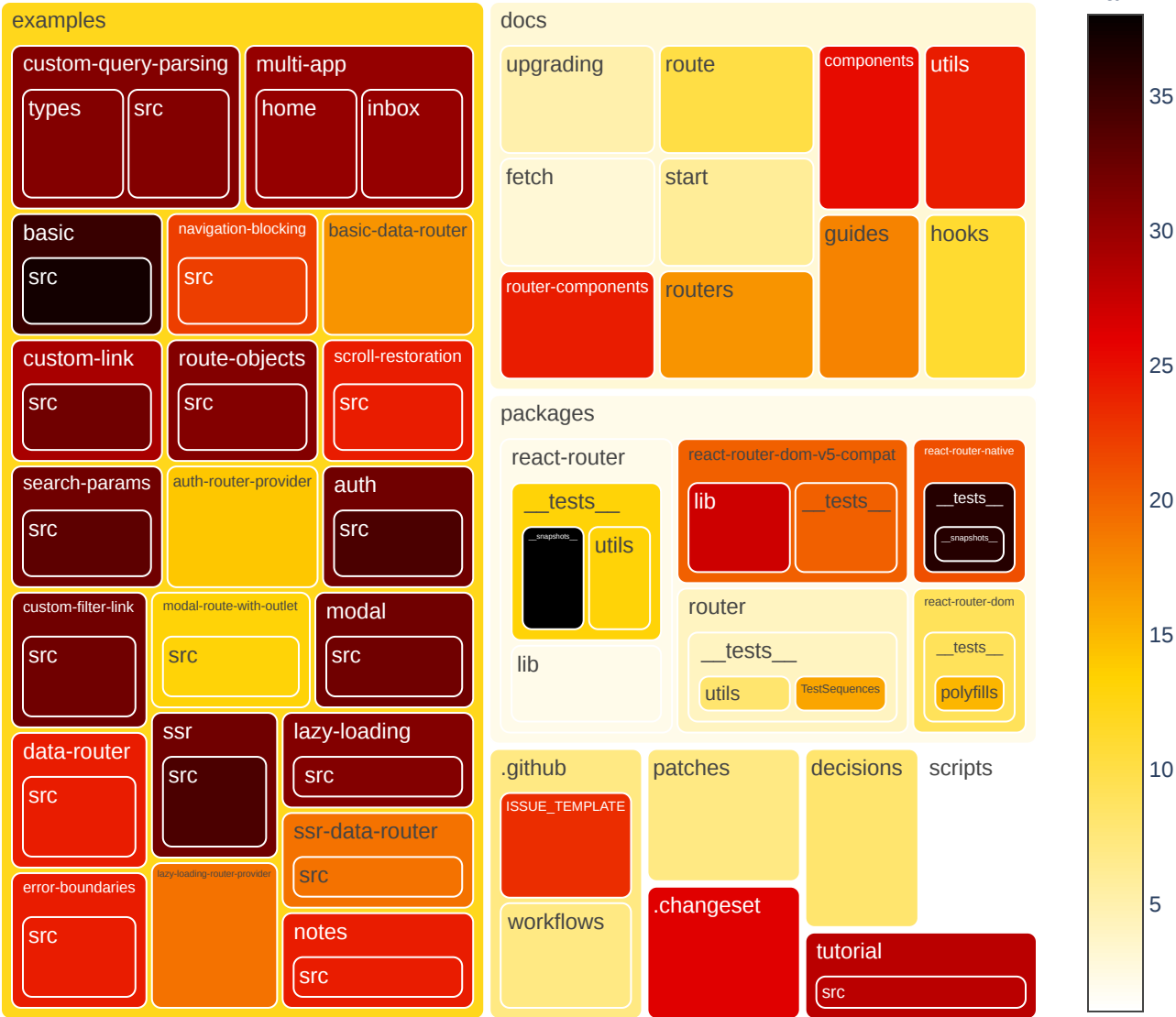
react-router-6.30.2



Days since last file creation per directory (ranked)

Rank of days since last file creation

react-router-6.30.2



Days since last file modification per directory

Days since last file modification

react-router-6.30.2



Days since last file modification per directory (ranked)

Rank of days since last file modification

react-router-6.30.2



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 229

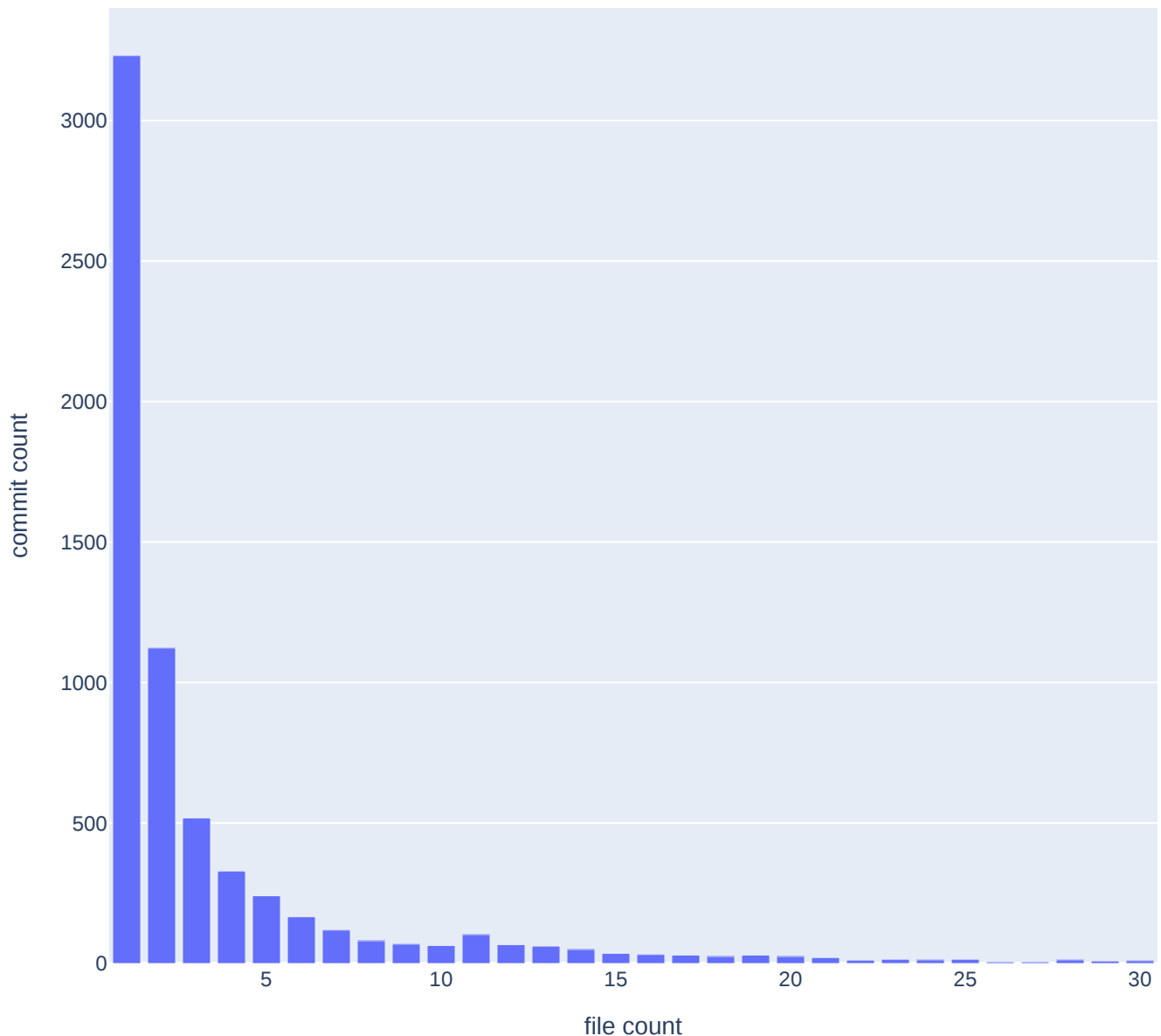
Max changed files with one commit = 498

	filesPerCommit	commitCount
count	117.000000	117.000000
mean	79.179487	57.692308
std	78.464424	319.376992
min	1.000000	1.000000
25%	30.000000	1.000000
50%	59.000000	3.000000
75%	96.000000	8.000000
max	498.000000	3231.000000

	filesPerCommit	commitCount
0	1	3231
1	2	1123
2	3	518
3	4	329
4	5	241
5	6	166
6	7	119
7	8	81
8	9	69
9	10	64
10	11	103
11	12	67
12	13	62
13	14	50
14	15	36
15	16	32
16	17	30
17	18	25
18	19	30
19	20	25
20	21	21
21	22	12
22	23	15
23	24	13
24	25	15
25	26	6
26	27	6
27	28	13
28	29	9
29	30	10

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files

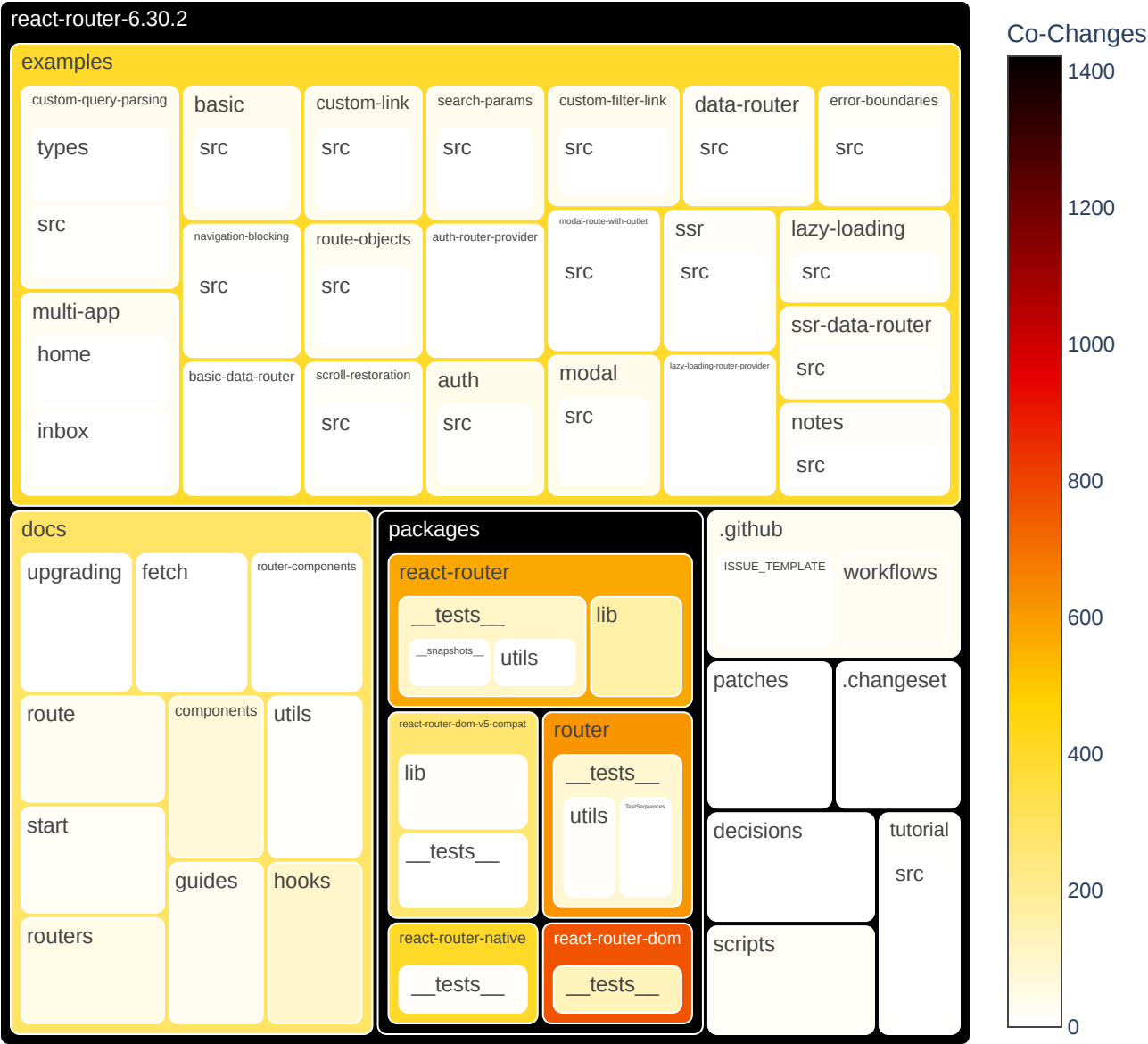
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

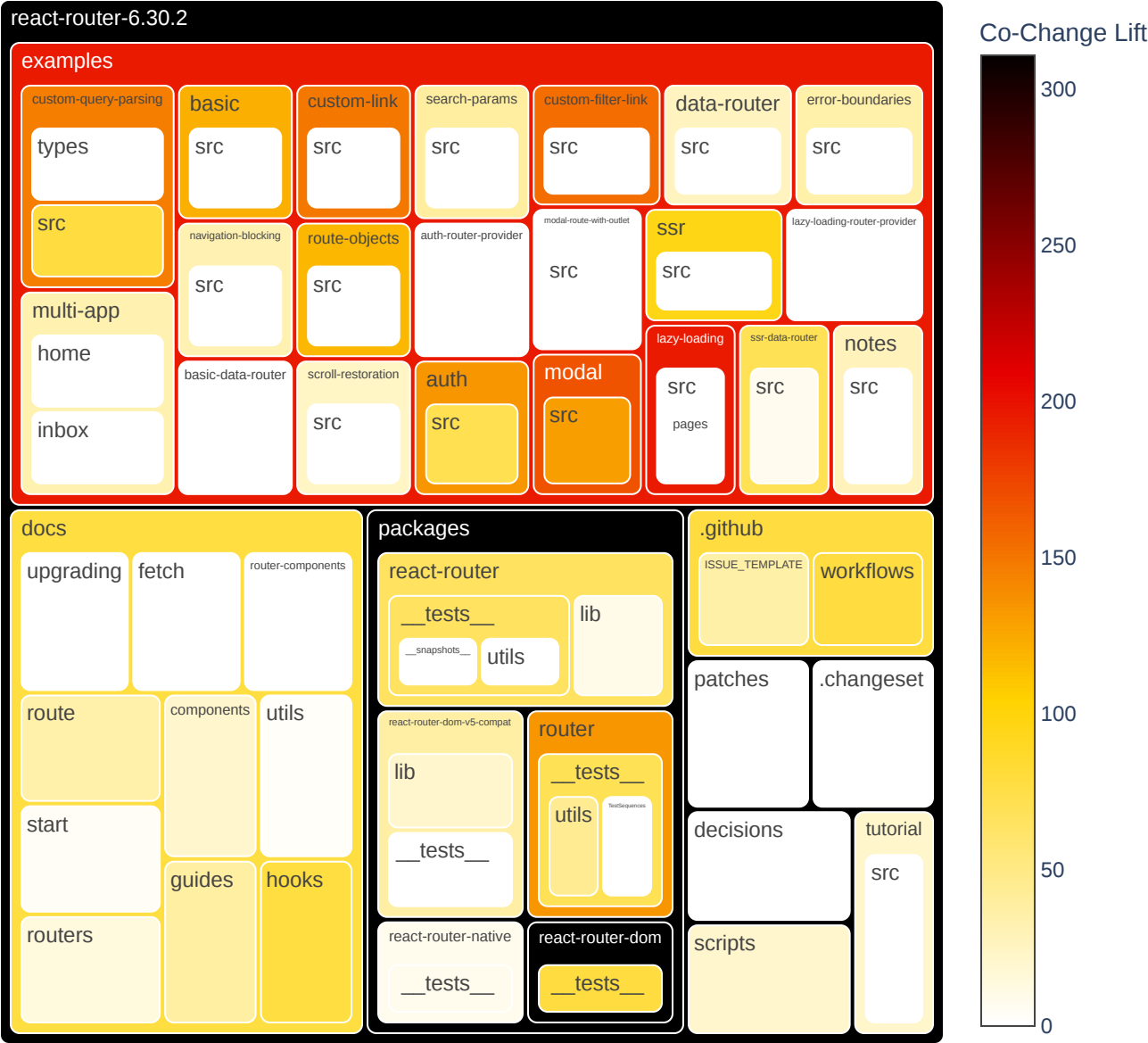
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

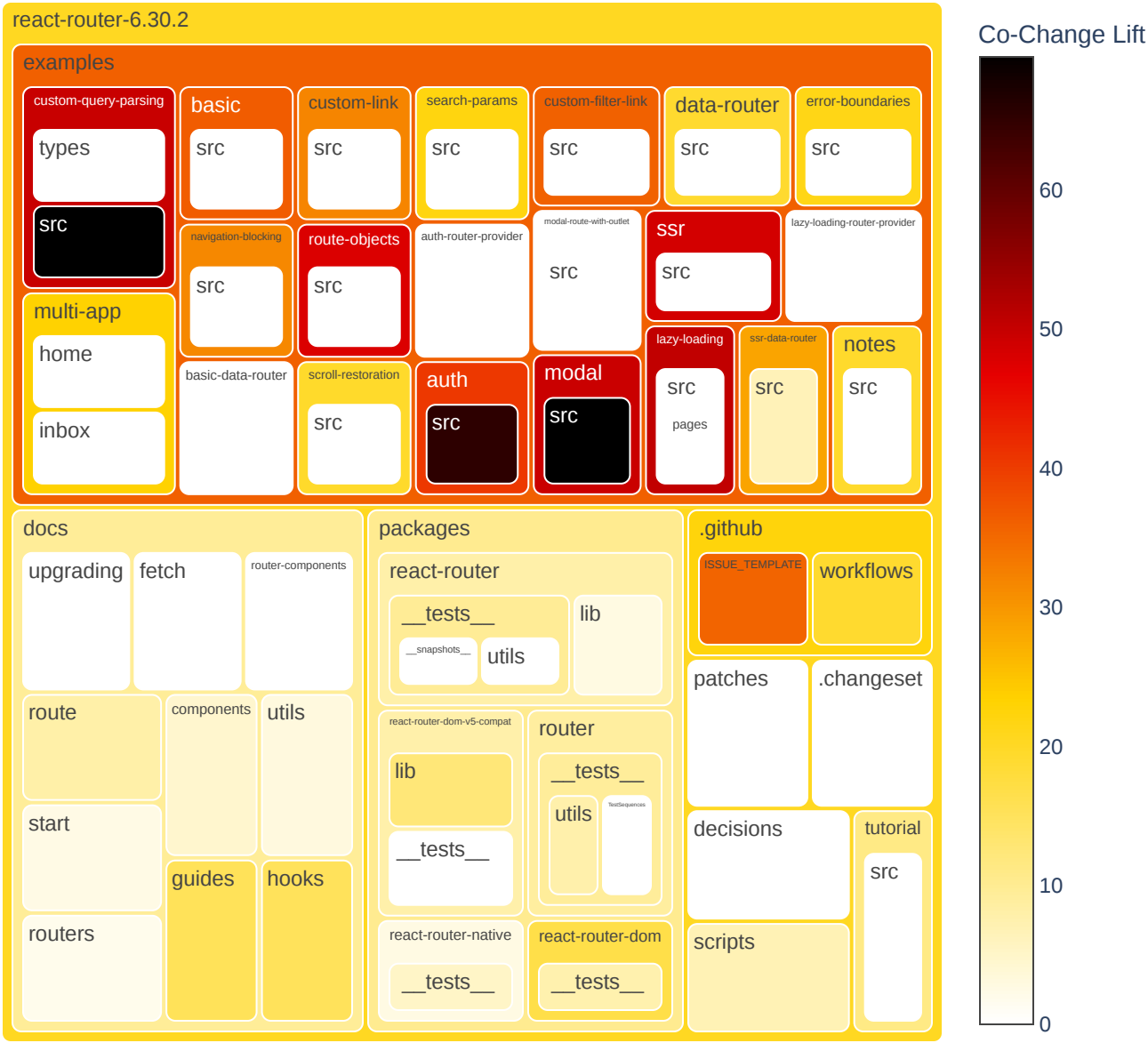
Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,



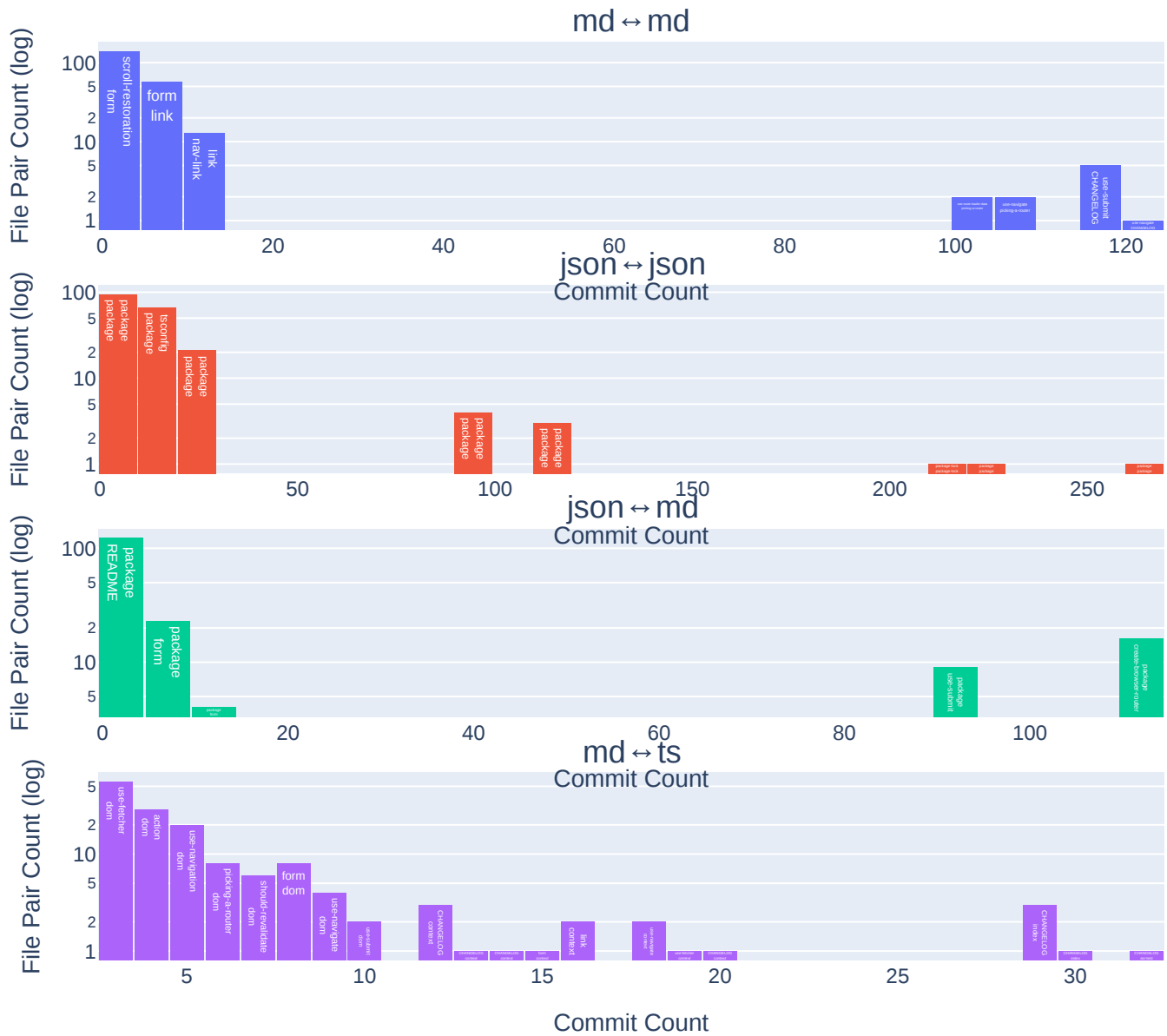
	fileExtensionPair	fileExtensionPairCount
0	md↔md	220
1	json↔json	192
2	json↔md	175
3	md↔ts	150

Files changed together by commit count

	fileExtensionPair	updateCommitCount	GroupRank	filePair	filePairWithRelativePath
0	md↔md	123	1	CHANGELOG↔CHANGELOG	packages/react-router-dom/CHANGELOG.md↔packages/react-router/CHANGELOG.md
1	md↔md	117	2	CHANGELOG↔CHANGELOG	packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/react-router-dom/CHANGELOG.md
2	md↔md	108	3	CHANGELOG↔CHANGELOG	packages/react-router/CHANGELOG.md↔packages/router/CHANGELOG.md
3	md↔md	106	4	CHANGELOG↔CHANGELOG	packages/react-router-dom/CHANGELOG.md↔packages/router/CHANGELOG.md
4	md↔md	100	5	CHANGELOG↔CHANGELOG	packages/react-router-native/CHANGELOG.md↔packages/router/CHANGELOG.md
5	md↔md	12	6	use-fetcher↔use-submit	docs/hooks/use-fetcher.md↔docs/hooks/use-submit.md
6	md↔md	11	7	link↔use-navigate	docs/components/link.md↔docs/hooks/use-navigate.md
7	md↔md	10	8	form↔use-navigate	docs/components/form.md↔docs/hooks/use-navigate.md
8	md↔md	9	9	form↔use-submit	docs/components/form.md↔docs/hooks/use-submit.md
9	md↔md	8	10	form↔link	docs/components/form.md↔docs/components/link.md
10	json↔json	264	1	package↔package	packages/react-router-dom/package.json↔packages/react-router/package.json
11	json↔json	220	2	package↔package	packages/react-router-dom/package.json↔packages/react-router-native/package.json
12	json↔json	215	3	package↔package	packages/react-router-native/package.json↔packages/react-router/package.json
13	json↔json	117	4	package↔package	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/package.json
14	json↔json	98	5	package↔package	packages/react-router-dom-v5-compat/package.json↔packages/router/package.json
15	json↔json	97	6	package↔package	packages/react-router-native/package.json↔packages/router/package.json
16	json↔json	21	7	package↔package	examples/auth/package.json↔examples/basic/package.json
17	json↔json	20	8	package↔package	examples/basic/package.json↔examples/custom-filter-link/package.json
18	json↔json	19	9	package↔package	packages/react-router-dom/package.json↔examples/auth/package.json
19	json↔json	18	10	package↔package	packages/react-router-native/package.json↔examples/custom-filter-link/package.json
20	json↔md	112	1	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/CHANGELOG.md
21	json↔md	111	2	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md
22	json↔md	94	3	package↔CHANGELOG	packages/router/package.json↔packages/react-router-dom/CHANGELOG.md
23	json↔md	93	4	package↔CHANGELOG	packages/router/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md
24	json↔md	11	5	package↔DEVELOPMENT	packages/react-router-dom-v5-compat/package.json↔DEVELOPMENT.md
25	json↔md	9	6	package↔create-browser-router	packages/react-router-dom/package.json↔docs/routers/create-browser-router.md
26	json↔md	8	7	package↔tutorial	packages/react-router/package.json↔docs/start/tutorial.md
27	json↔md	6	8	package↔use-navigate	packages/react-router/package.json↔docs/hooks/use-navigate.md
28	json↔md	5	9	package↔link	packages/react-router-dom-v5-compat/package.json↔docs/components/link.md
29	json↔md	4	10	package↔form	packages/react-router-native/package.json↔docs/components/form.md
30	md↔ts	32	1	CHANGELOG↔router	packages/react-router/CHANGELOG.md↔packages/router/router.ts
31	md↔ts	30	2	CHANGELOG↔router	packages/router/CHANGELOG.md↔packages/router/router.ts
32	md↔ts	29	3	CHANGELOG↔router	packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/router/router.ts
33	md↔ts	20	4	CHANGELOG↔utils	packages/react-router/CHANGELOG.md↔packages/router/utils.ts
34	md↔ts	19	5	CHANGELOG↔utils	packages/router/CHANGELOG.md↔packages/router/utils.ts
35	md↔ts	18	6	CHANGELOG↔utils	packages/react-router-dom/CHANGELOG.md↔packages/router/utils.ts
36	md↔ts	16	7	CHANGELOG↔utils	packages/react-router-dom-v5-compat/CHANGELOG.md↔packages/router/utils.ts
37	md↔ts	15	8	form↔router	docs/components/form.md↔packages/router/router.ts

	fileExtensionPair	updateCommitCount	GroupRank	filePair	filePairWithRelativePath
38	md↔ts	14	9	use-submit↔router	docs/hooks/use-submit.md↔packages/router/router.ts
39	md↔ts	13	10	use-submit↔dom	docs/hooks/use-submit.md↔packages/react-router-dom/dom.ts

Co-Changed Files by their commit count

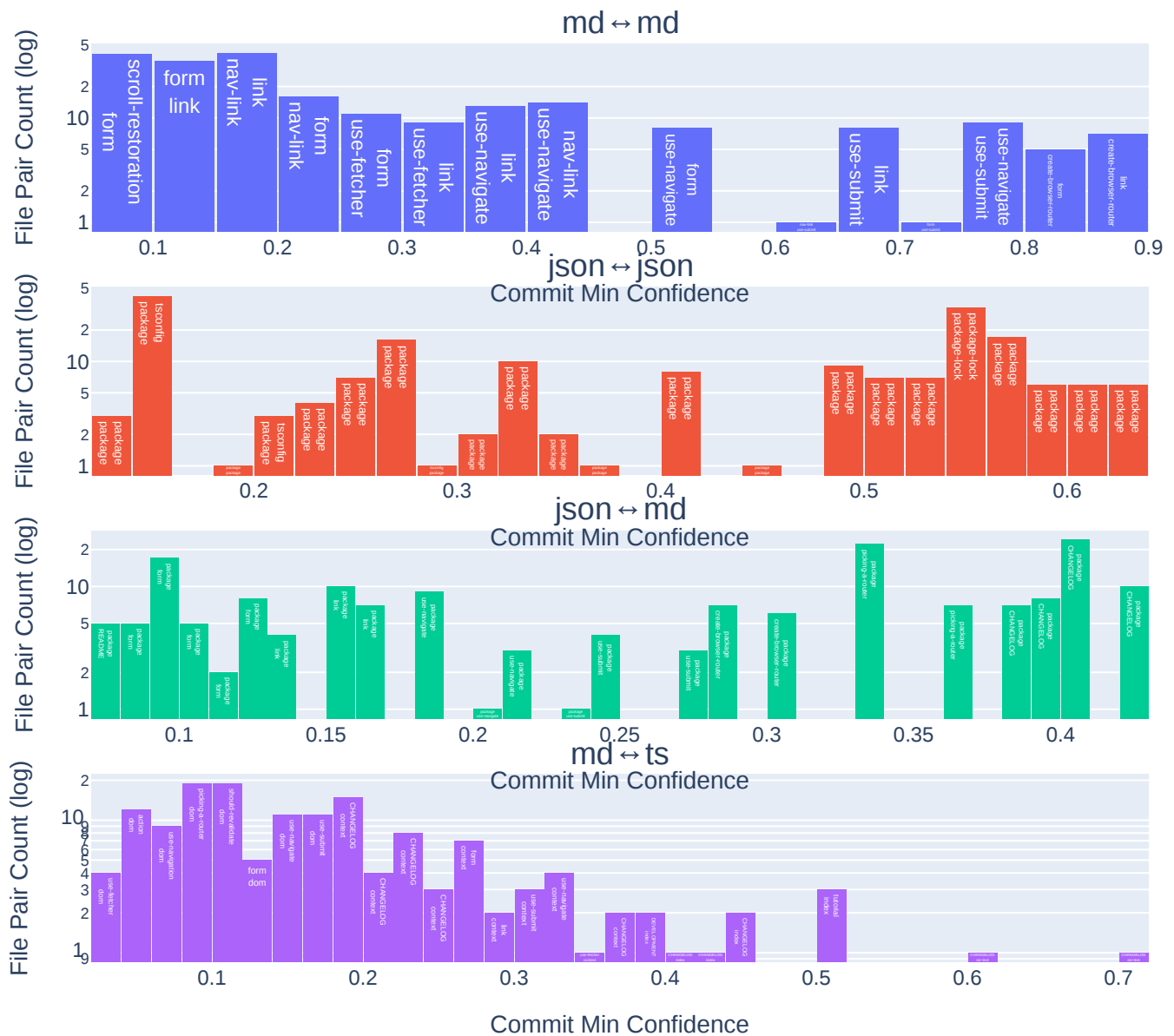


Files changed together by commit min confidence

The commit min confidence is the commit count where both files were changed divided by the commit count of the file with the least commits. This metric is useful to identify pairs of files that are frequently changed together and is not biased by single files that are changed very often.

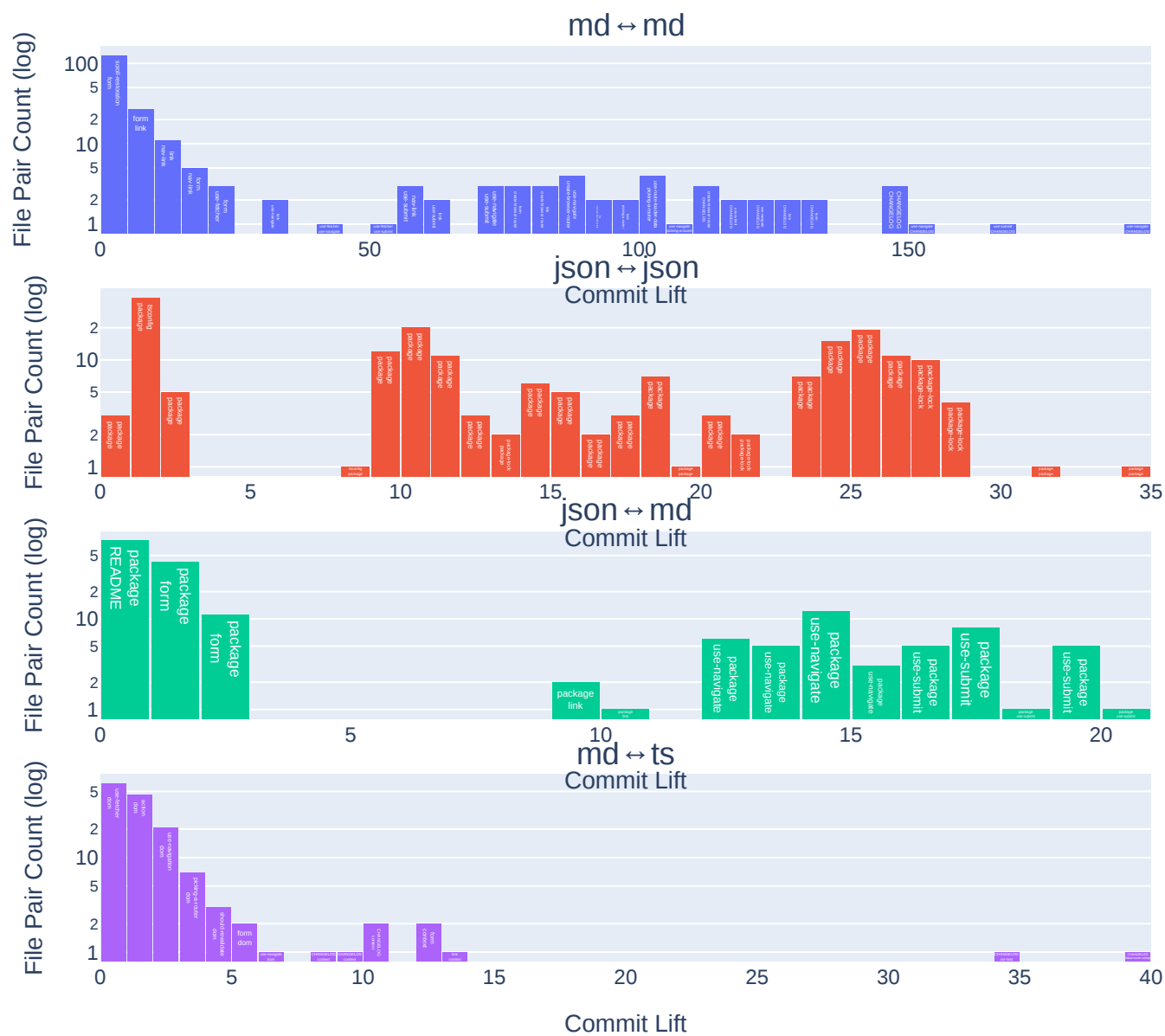
	fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair	filePairWithRelativePath
0	md↔md	0.888889	1	README↔README	examples/custom-filter-link/README.md↔examples/custom-link/README.md
1	md↔md	0.857143	2	README↔README	examples/basic/README.md↔examples/modal/README.md
2	md↔md	0.800000	3	README↔README	examples/ssr/README.md↔examples/auth/README.md
3	md↔md	0.750000	4	README↔README	examples/ssr/README.md↔examples/route-objects/README.md
4	md↔md	0.727273	5	README↔README	examples/ssr/README.md↔examples/basic/README.md
5	md↔md	0.666667	6	README↔README	examples/ssr/README.md↔examples/custom-filter-link/README.md
6	md↔md	0.600000	7	link↔use-view-transition-state	docs/components/link.md↔docs/hooks/use-view-transition-state.md
7	md↔md	0.500000	8	README↔README	examples/custom-link/README.md↔examples/lazy-loading/README.md
8	md↔md	0.446281	9	CHANGELOG↔CHANGELOG	packages/react-router/CHANGELOG.md↔packages/router/CHANGELOG.md
9	md↔md	0.444444	10	use-route-loader-data↔picking-a-router	docs/hooks/use-route-loader-data.md↔docs/routers/picking-a-router.md
10	json↔json	0.625000	1	package↔package	examples/basic/package.json↔examples/custom-filter-link/package.json
11	json↔json	0.600000	2	package↔package	examples/auth/package.json↔examples/basic/package.json
12	json↔json	0.588235	3	package↔package	examples/basic/package.json↔examples/custom-link/package.json
13	json↔json	0.581498	4	package↔package	packages/react-router-dom/package.json↔packages/react-router/package.json
14	json↔json	0.571429	5	package↔package	examples/basic/package.json↔examples/modal/package.json
15	json↔json	0.566667	6	package↔package	examples/custom-query-parsing/package.json↔examples/multi-app/package.json
16	json↔json	0.562500	7	package↔package	packages/react-router-native/package.json↔examples/custom-filter-link/package.json
17	json↔json	0.548387	8	package↔package	examples/basic/package.json↔examples/custom-query-parsing/package.json
18	json↔json	0.542857	9	package↔package	packages/react-router-dom/package.json↔examples/auth/package.json
19	json↔json	0.529412	10	package↔package	packages/react-router/package.json↔examples/custom-link/package.json
20	json↔md	0.428571	1	package↔README	examples/auth/package.json↔examples/modal/README.md
21	json↔md	0.400722	2	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md
22	json↔md	0.400000	3	package↔CHANGELOG	packages/router/package.json↔packages/react-router-dom/CHANGELOG.md
23	json↔md	0.395745	4	package↔CHANGELOG	packages/router/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.md
24	json↔md	0.392982	5	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/CHANGELOG.md
25	json↔md	0.391608	6	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router/CHANGELOG.md
26	json↔md	0.388430	7	package↔CHANGELOG	packages/react-router-native/package.json↔packages/router/CHANGELOG.md
27	json↔md	0.386207	8	package↔CHANGELOG	packages/react-router-dom/package.json↔packages/react-router/CHANGELOG.md
28	json↔md	0.363636	9	package↔README	examples/auth/package.json↔examples/basic/README.md
29	json↔md	0.333333	10	package↔use-route-loader-data	packages/react-router-dom-v5-compat/package.json↔docs/hooks/use-route-loader-data.md
30	md↔ts	0.714286	1	create-static-handler↔router	docs/routers/create-static-handler.md↔packages/router/router.ts
31	md↔ts	0.600000	2	should-revalidate↔should-revalidate-test	docs/route/should-revalidate.md↔packages/router/__tests__/should-revalidate-test.ts
32	md↔ts	0.500000	3	README↔router	examples/ssr-data-router/README.md↔packages/router/router.ts
33	md↔ts	0.444444	4	use-route-loader-data↔utils	docs/hooks/use-route-loader-data.md↔packages/router/utils.ts
34	md↔ts	0.428571	5	create-static-handler↔ssr-test	docs/routers/create-static-handler.md↔packages/router/__tests__/ssr-test.ts
35	md↔ts	0.409091	6	api-development-strategy↔router	docs/guides/api-development-strategy.md↔packages/router/router.ts

fileExtensionPair	updateCommitMinConfidence	GroupRank	filePair	filePairWithRelativePath
36	md↔ts	0.388889	7	router↔create-browser-router↔lazy-discovery-test docs/routers/create-browser-router.md↔packages/router/_tests_/lazy-discovery-test.ts
37	md↔ts	0.384615	8	README↔router packages/router/README.md↔packages/router/router.ts
38	md↔ts	0.375000	9	use-fetcher↔router docs/hooks/use-fetcher.md↔packages/router/router.ts
39	md↔ts	0.370370	10	should-revalidate↔router docs/route/should-revalidate.md↔packages/router/router.ts



	fileExtensionPair	updateCommitLift	GroupRank	filePair	filePairWithRelativePath
0	md↔md	194.250000	1	README↔README	examples/lazy-loading/README.md↔examples/route-objects/README.md
1	md↔md	166.500000	2	README↔README	examples/modal/README.md↔examples/route-objects/README.md
2	md↔md	153.481481	3	README↔README	examples/custom-filter-link/README.md↔examples/custom-link/README.md
3	md↔md	148.000000	4	README↔README	examples/custom-link/README.md↔examples/modal/README.md
4	md↔md	145.687500	5	README↔README	examples/custom-query-parsing/README.md↔examples/route-objects/README.md
5	md↔md	133.200000	6	README↔README	examples/auth/README.md↔examples/modal/README.md
6	md↔md	129.500000	7	README↔README	examples/custom-link/README.md↔examples/route-objects/README.md
7	md↔md	124.320000	8	README↔README	examples/auth/README.md↔examples/search-params/README.md
8	md↔md	121.090909	9	README↔README	examples/basic/README.md↔examples/modal/README.md
9	md↔md	116.550000	10	README↔README	examples/auth/README.md↔examples/route-objects/README.md
10	json↔json	34.533333	1	package↔package	examples/error-boundaries/package.json↔examples/notes/package.json
11	json↔json	31.393939	2	package-lock↔package	examples/navigation-blocking/package-lock.json↔examples/navigation-blocking/package.json
12	json↔json	28.566176	3	package↔package	examples/custom-filter-link/package.json↔examples/custom-link/package.json
13	json↔json	28.406452	4	package↔package	examples/custom-query-parsing/package.json↔examples/multi-app/package.json
14	json↔json	27.750000	5	package↔package	examples/auth/package.json↔examples/custom-filter-link/package.json
15	json↔json	27.626667	6	package↔package	examples/notes/package.json↔examples/scroll-restoration/package.json
16	json↔json	27.518750	7	package↔package	examples/custom-filter-link/package.json↔examples/multi-app/package.json
17	json↔json	27.490114	8	package↔package	examples/custom-query-parsing/package.json↔examples/lazy-loading/package.json
18	json↔json	26.979167	9	package↔package	examples/basic/package.json↔examples/custom-filter-link/package.json
19	json↔json	26.640000	10	package↔package	examples/auth/package.json↔examples/search-params/package.json
20	json↔md	20.812500	1	package↔README	examples/custom-filter-link/package.json↔examples/modal/README.md
21	json↔md	19.588235	2	package↔README	examples/custom-link/package.json↔examples/modal/README.md
22	json↔md	19.028571	3	package↔README	examples/auth/package.json↔examples/modal/README.md
23	json↔md	18.500000	4	package↔README	examples/basic/package.json↔examples/modal/README.md
24	json↔md	17.760000	5	package↔README	examples/auth/package.json↔examples/auth/README.md
25	json↔md	17.266667	6	package↔README	examples/basic/package.json↔examples/auth/README.md
26	json↔md	16.187500	7	package↔README	examples/custom-filter-link/package.json↔examples/custom-filter-link/README.md
27	json↔md	16.145455	8	package↔README	examples/auth/package.json↔examples/basic/README.md
28	json↔md	15.696970	9	package↔README	examples/basic/package.json↔examples/basic/README.md
29	json↔md	15.235294	10	package↔README	examples/custom-link/package.json↔examples/custom-filter-link/README.md
30	md↔ts	39.176471	1	create-static-handler↔ssr-test	docs/routers/create-static-handler.md↔packages/router/__tests__/ssr-test.ts
31	md↔ts	34.533333	2	should-revalidate↔should-revalidate-test	docs/route/should-revalidate.md↔packages/router/__tests__/should-revalidate-test.ts
32	md↔ts	13.875000	3	create-browser-router↔lazy-test	docs/routers/create-browser-router.md↔packages/router/__tests__/lazy-test.ts
33	md↔ts	12.465241	4	api-development-strategy↔ssr-test	docs/guides/api-development-strategy.md↔packages/router/__tests__/ssr-test.ts
34	md↔ts	12.046512	5	use-route-loader-data↔history	docs/hooks/use-route-loader-data.md↔packages/router/history.ts
35	md↔ts	10.791667	6	create-browser-router↔lazy-discovery-test	docs/routers/create-browser-router.md↔packages/router/__tests__/lazy-discovery-test.ts
36	md↔ts	10.791667	7	should-revalidate↔fetchers-test	docs/route/should-revalidate.md↔packages/router/__tests__/fetchers-test.ts
37	md↔ts	9.296825	8	use-submit↔dom	docs/hooks/use-submit.md↔packages/react-router-dom/dom.ts
38	md↔ts	8.246462	9	use-fetcher↔dom	docs/hooks/use-fetcher.md↔packages/react-router-dom/dom.ts
39	md↔ts	6.937500	10	create-browser-router↔fetchers-test	docs/routers/create-browser-router.md↔packages/router/__tests__/fetchers-test.ts

Co-Changed Files by their commit lift

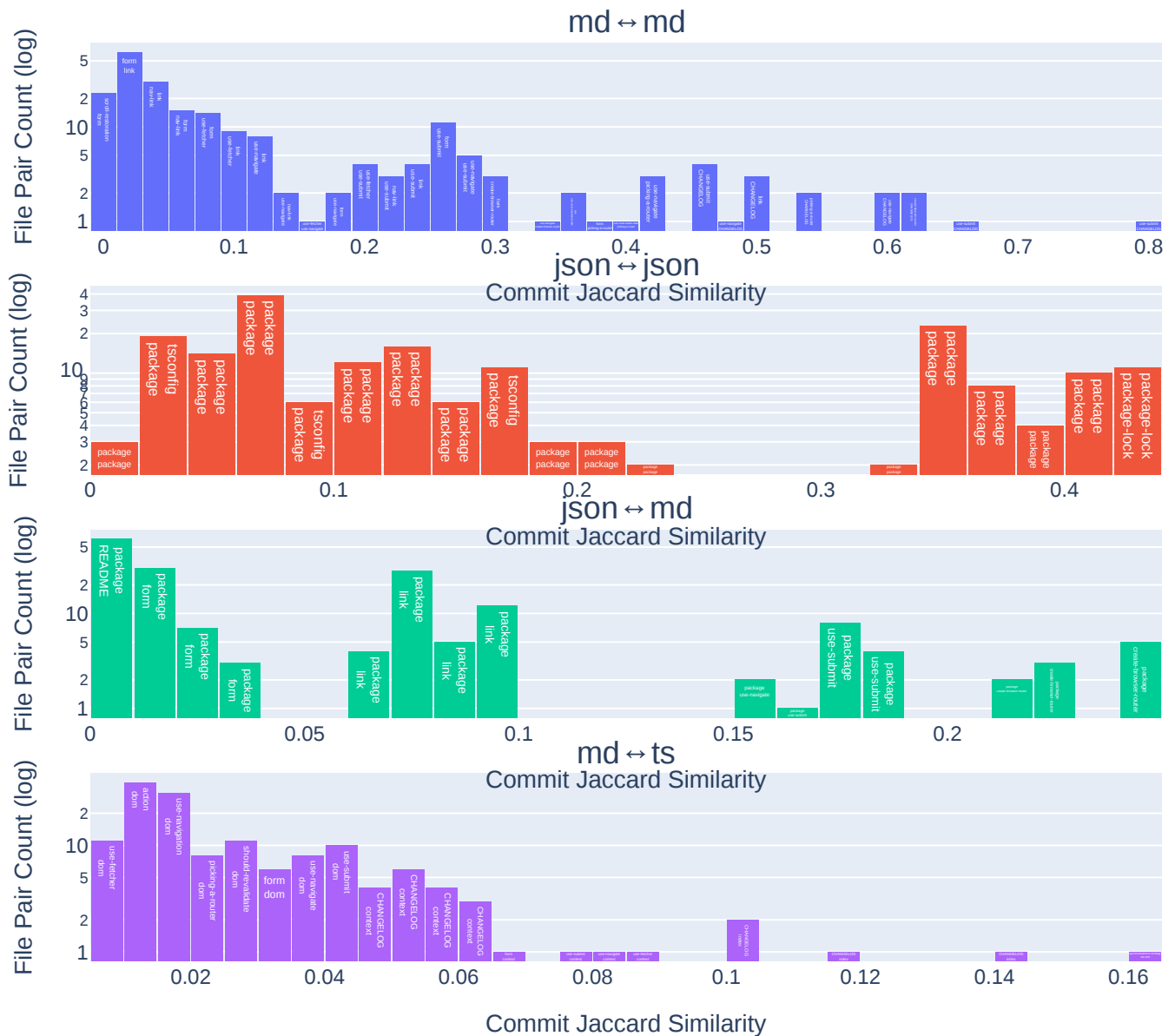


Files changed together by commit Jaccard similarity

	fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	filePair	filePairWithRelativePath
0	md↔md	0.800000	1	README↔README	examples/custom-filter-link/README.md↔examples/custom-link/README.mc
1	md↔md	0.666667	2	README↔README	examples/auth/README.md↔examples/search-params/README.mc
2	md↔md	0.615385	3	README↔README	examples/auth/README.md↔examples/basic/README.mc
3	md↔md	0.600000	4	README↔README	examples/custom-link/README.md↔examples/modal/README.mc
4	md↔md	0.545455	5	README↔README	examples/auth/README.md↔examples/modal/README.mc
5	md↔md	0.500000	6	README↔README	examples/ssr/README.md↔examples/auth/README.mc
6	md↔md	0.470588	7	README↔README	examples/ssr/README.md↔examples/basic/README.mc
7	md↔md	0.461538	8	README↔README	examples/auth/README.md↔examples/custom-filter-link/README.mc
8	md↔md	0.428571	9	README↔README	examples/basic/README.md↔examples/custom-filter-link/README.mc
9	md↔md	0.400000	10	README↔README	examples/ssr/README.md↔examples/modal/README.mc
10	json↔json	0.434783	1	package↔package	examples/custom-filter-link/package.json↔examples/custom-link/package.jsor
11	json↔json	0.428571	2	package↔package	examples/auth/package.json↔examples/search-params/package.jsor
12	json↔json	0.425532	3	package↔package	examples/auth/package.json↔examples/custom-filter-link/package.jsor
13	json↔json	0.420000	4	package↔package	examples/auth/package.json↔examples/basic/package.jsor
14	json↔json	0.416667	5	package↔package	examples/basic/package.json↔examples/custom-filter-link/package.jsor
15	json↔json	0.408163	6	package↔package	examples/auth/package.json↔examples/custom-link/package.jsor
16	json↔json	0.406780	7	package↔package	packages/react-router-dom/package.json↔packages/react-router/package.jsor
17	json↔json	0.400000	8	package↔package	examples/basic/package.json↔examples/custom-link/package.jsor
18	json↔json	0.392157	9	package↔package	examples/basic/package.json↔examples/modal/package.jsor
19	json↔json	0.386364	10	package↔package	examples/custom-query-parsing/package.json↔examples/multi-app/package.jsor
20	json↔md	0.245575	1	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.mc
21	json↔md	0.244009	2	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router-dom/CHANGELOG.mc
22	json↔md	0.242188	3	package↔CHANGELOG	packages/router/package.json↔packages/router/CHANGELOG.mc
23	json↔md	0.241379	4	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/react-router/CHANGELOG.mc
24	json↔md	0.221957	5	package↔CHANGELOG	packages/router/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.mc
25	json↔md	0.220657	6	package↔CHANGELOG	packages/router/package.json↔packages/react-router-dom/CHANGELOG.mc
26	json↔md	0.218097	7	package↔CHANGELOG	packages/router/package.json↔packages/react-router/CHANGELOG.mc
27	json↔md	0.216590	8	package↔CHANGELOG	packages/react-router-dom-v5-compat/package.json↔packages/router/CHANGELOG.mc
28	json↔md	0.189097	9	package↔CHANGELOG	packages/react-router-native/package.json↔packages/react-router-dom-v5-compat/CHANGELOG.mc
29	json↔md	0.188552	10	package↔CHANGELOG	packages/react-router-native/package.json↔packages/react-router-dom/CHANGELOG.mc
30	md↔ts	0.160494	1	use-submit↔dom	docs/hooks/use-submit.md↔packages/react-router-dom/dom.ts
31	md↔ts	0.142857	2	create-static-handler↔ssr-test	docs/routers/create-static-handler.md↔packages/router/__tests__/ssr-test.ts
32	md↔ts	0.118421	3	use-fetcher↔dom	docs/hooks/use-fetcher.md↔packages/react-router-dom/dom.ts
33	md↔ts	0.104478	4	create-browser-router↔lazy-discovery-test	docs/routers/create-browser-router.md↔packages/router/__tests__/lazy-discovery-test.ts
34	md↔ts	0.103448	5	should-revalidate↔should-revalidate-test	docs/route/should-revalidate.md↔packages/router/__tests__/should-revalidate-test.ts
35	md↔ts	0.088889	6	form↔dom	docs/components/form.md↔packages/react-router-dom/dom.ts
36	md↔ts	0.083333	7	api-development-strategy↔ssr-test	docs/guides/api-development-strategy.md↔packages/router/__tests__/ssr-test.ts

	fileExtensionPair	updateCommitJaccardSimilarity	GroupRank	filePair	filePairWithRelativePath
37	md↔ts	0.075000	8	should-revalidate↔fetchers-test	docs/route/should-revalidate.md↔packages/router/___tests___/fetchers-test.ts
38	md↔ts	0.068376	9	create-browser-router↔index	docs/routers/create-browser-router.md↔packages/router/index.ts
39	md↔ts	0.061224	10	use-route-loader-data↔history	docs/hooks/use-route-loader-data.md↔packages/router/history.ts

Co-Changed Files by their commit jaccard similarity



Find pairwise changed files with many highly ranked metrics

Find those pairwise changed files that have a high rank in many metrics by calculating a combined (weighted) score based on the ranks of each metric. This is useful to identify pairs of files that score high in most metrics, which indicates a strong co-change relationship.

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountExtensionRank	updateCommitMinConfidenceExtensionRa
0	json↔json	package↔package	13	8	
1	json↔json	package↔package	17	8	
2	json↔json	package↔package	17	8	
3	json↔json	package↔package	17	8	
4	json↔json	package↔package	17	8	
5	json↔json	package↔package	21	7	
6	json↔json	package↔package	21	7	
7	json↔json	package↔package	21	7	
8	json↔json	package↔package	23	8	
9	json↔json	package↔package	27	7	
10	json↔md	package↔CHANGELOG	33	2	
11	json↔md	package↔CHANGELOG	33	2	
12	json↔md	package↔CHANGELOG	35	4	
13	json↔md	package↔CHANGELOG	38	4	
14	json↔md	package↔CHANGELOG	38	1	
15	json↔md	package↔CHANGELOG	38	4	
16	json↔md	package↔README	38	10	
17	json↔md	package↔README	38	10	
18	json↔md	package↔README	38	10	
19	json↔md	package↔README	38	11	
20	md↔md	README↔README	15	10	
21	md↔md	README↔README	22	12	
22	md↔md	README↔README	22	12	
23	md↔md	README↔README	23	10	
24	md↔md	README↔README	25	12	
25	md↔md	README↔README	25	12	
26	md↔md	README↔README	27	10	
27	md↔md	README↔README	27	10	
28	md↔md	README↔README	29	12	
29	md↔md	README↔README	29	15	
30	md↔ts	create-static-handler↔ssr-test	27	19	
31	md↔ts	should-revalidate↔should-revalidate-test	28	19	
32	md↔ts	create-browser-router↔lazy-discovery-test	32	15	
33	md↔ts	use-submit↔dom	33	10	
34	md↔ts	use-fetcher↔dom	42	13	
35	md↔ts	create-browser-router↔lazy-test	46	19	
36	md↔ts	use-route-loader-data↔history	46	19	

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCountExtensionRank	updateCommitMinConfidenceExtensionRa
37	md↔ts	create-browser-router↔fetchers-test	60	18	
38	md↔ts	api-development-strategy↔ssr-test	62	19	
39	md↔ts	should-revalidate↔fetchers-test	62	19	

Pairwise changed files with pareto-optimal metrics

A pair (count, confidence, jaccard, lift) is Pareto-optimal if there is no other pair that is better or equal in all metrics and strictly better in at least one. In other words, it is not "dominated" by any other pair.

The frontier = the “best tradeoffs.”

Pairwise changed files with pareto-optimal metrics - not considering file extensions

	filePair	combinedMetricsScore	updateCommitCount	updateCommitMinConfidence	updateCommitJaccardSimilarity	update
0	CHANGELOG↔CHANGELOG	131	117	0.422383	0.267735	
1	CHANGELOG↔CHANGELOG	130	123	0.431579	0.272124	
2	package↔package	74	264	0.581498	0.406780	
3	CHANGELOG↔CHANGELOG	128	108	0.446281	0.254717	
4	package↔CHANGELOG	35	93	0.395745	0.242188	
5	package↔package	13	20	0.625000	0.434783	
6	package↔package	21	21	0.600000	0.428571	
7	package↔package	21	21	0.600000	0.428571	
8	package↔package	21	21	0.600000	0.428571	
9	README↔README	15	8	0.888889	0.800000	
10	README↔README	29	3	0.750000	0.428571	

Pairwise changed files with pareto-optimal metrics - using ranks grouped by file extensions

	fileExtensionPair	filePair	combinedMetricsScore	updateCommitCount	updateCommitMinConfidence	updateCommitJaccardSimila
0	md↔ts	CHANGELOG↔context	353	3	0.053571	0.000
1	md↔md	submit↔use-CHANGELOG	424	3	0.073171	0.000
2	md↔ts	CHANGELOG↔router-test	332	6	0.033898	0.010

WordCloud of git authors

	word	frequency
0	Michael Jackson	1832
1	Ryan Florence	1119
2	Matt Brophy	772
3	Jimmy Jia	381
4	Tim Dorr	326
5	Chance Strickland	180
6	Mateusz Zatorski	124
7	Logan McAnsh	99
8	Dan Abramov	92
9	Paul Sherman	77

